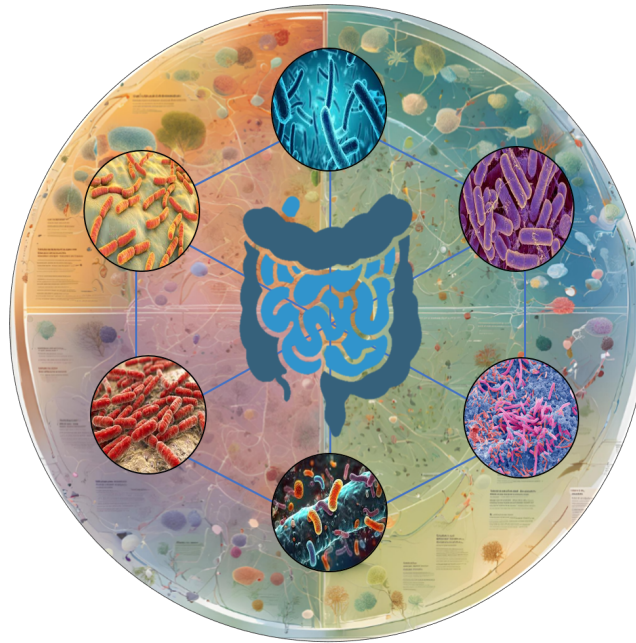


## PHENOME INDIA: CSIR COHORT HEALTH KNOWLEDGEBASE



# Gut Microbiome Report

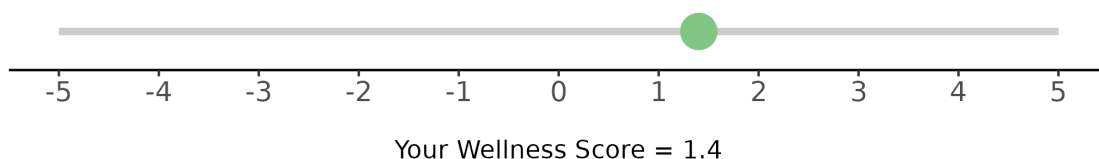
Lab ID: 0BPXYV

Date of sample collection: 4/5/2024

**Disclaimer:** These are test results from a research level testing protocol and do not necessarily indicate clinical results done at accredited hospital or lab environments. These are for your reference purposes only and should not be used to guide any therapy or treatment. If you have any doubts regarding your results, please contact your physician. These reports and results cannot be used for any medico-legal liability and your consent to participation is to accept these results in the form as it is whatsoever without any liability to CSIR, participating labs, IGIB and its lab members, project Principal Investigator and Co-Investigators and associated research staff.

## Gut Microbiome Wellness Index 2 (GMWI2):

GMWI2 is a standardized disease-agnostic health status indicator based on gut microbiome taxonomic profiles. A positive GMWI2 value indicates a greater abundance of healthy, beneficial taxa, while a negative GMWI2 value reflects a higher abundance of less beneficial taxa. When the GMWI2 value is 0, it shows a balanced presence of beneficial and less beneficial taxa, categorizing the sample as neutral in terms of health status. (Chang et al., "Gut Microbiome Wellness Index 2 enhances health status prediction from gut microbiome taxonomic profiles", *Nature Communications*, 2024, Vol. 15, Issue 1, pp. 7447)



Your gut microbiome wellness score is 1.4 which indicates good gut microbiome.

### Interpretation of Your Results:

Your overall gut microbiome analysis indicates a well-balanced and healthy microbiome, supporting optimal gut health.

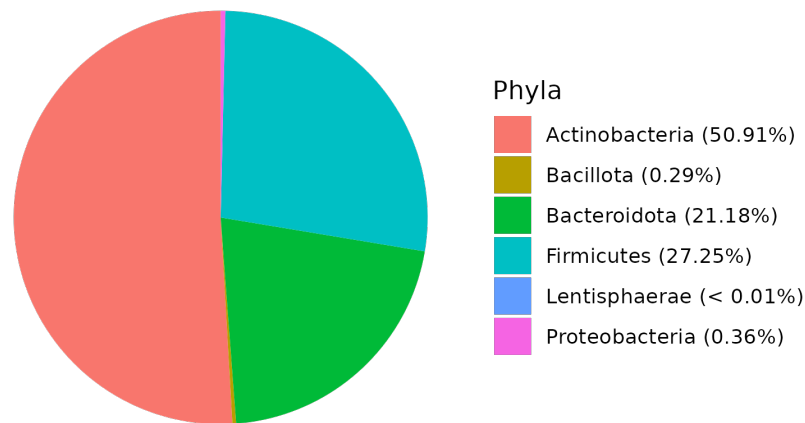
## Detailed Gut Microbiome Analysis

This report is the analysis of the bacterial diversity and abundance of different species present in the gut microbiome using stool sample for metagenomic profiling. Through this report we aim to offer valuable insights, empowering you to gain a deeper understanding of your gut health. From the stool sample collected from you by the cohort team, we isolated metagenomic DNA, which is sequenced using Next-generation sequencing (NGS) technology. The sequence data was analyzed using different bioinformatics pipelines to profile the microbial community in the sample. This information was used to generate your gut microbiome report. We aspire to equip you with insights that can guide you towards making informed decisions about your health and lifestyle. We have reported your results in the following sections:

1. Phylum Diversity
2. Firmicutes / Bacteroidetes Ratio
3. Species-level Diversity
  - 3.1 Species Richness
  - 3.2 Evenness Index
  - 3.3 Microbiome Diversity Index
  - 3.4 Top 15 Species
4. Probiotic Species and Short Chain Fatty Acid (SCFA) Producers

## Phylum-Level Diversity of Your Gut Microbiome:

The phylum diversity represents the variety of bacterial phyla present in your gut microbiome. Bacterial phyla are broad categories that contain different groups of bacteria sharing certain characteristics. In your microbiome report, we analyzed the number and relative abundance of various bacterial phyla present in your gut.

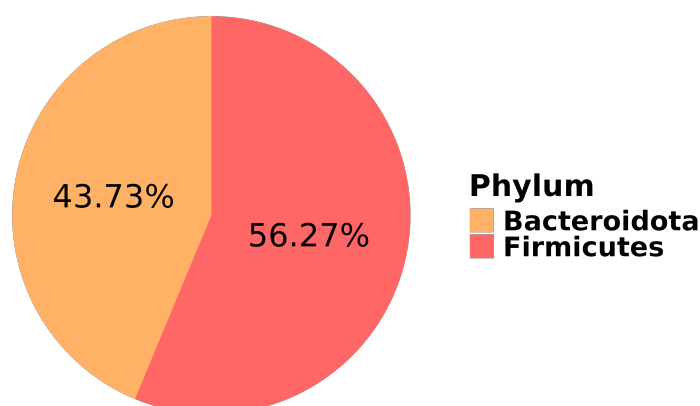


Your gut microbiome profile indicates the presence of 6 distinct phyla. The average range of phyla present in gut microbiome of Indian population is 3-15.\*

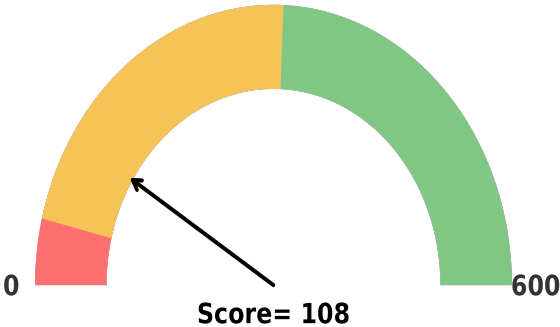
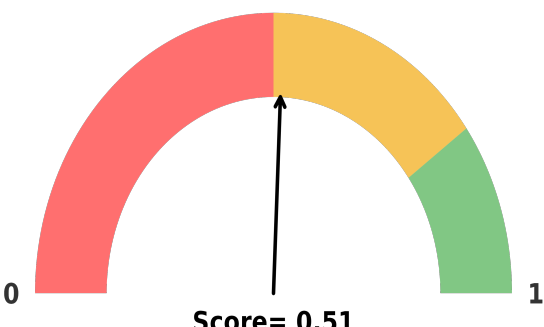
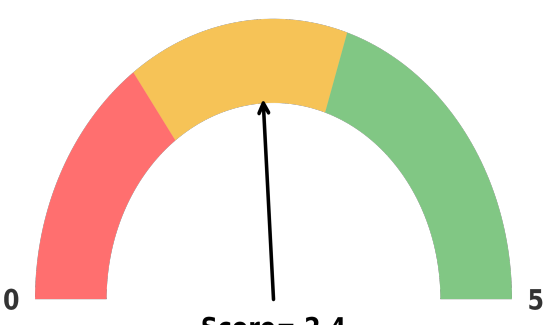
## Firmicutes and Bacteroidetes Ratio:

Firmicutes (F) and Bacteroidetes (B) are the most common bacterial phyla found in the gut microbiome. The ratio between the abundance of Firmicutes and Bacteroidetes, known as the F/B ratio, is used to measure the relative proportion of these organisms. A balanced proportion (F/B ratio = 1) of these two phyla is vital for maintaining good health, while the majority of the Indian population typically falls within an F/B ratio range of approximately 0.69 to 4.41.\*

$$F / B = 1.29$$



## Species-level Diversity of Your Gut Microbiome\*:

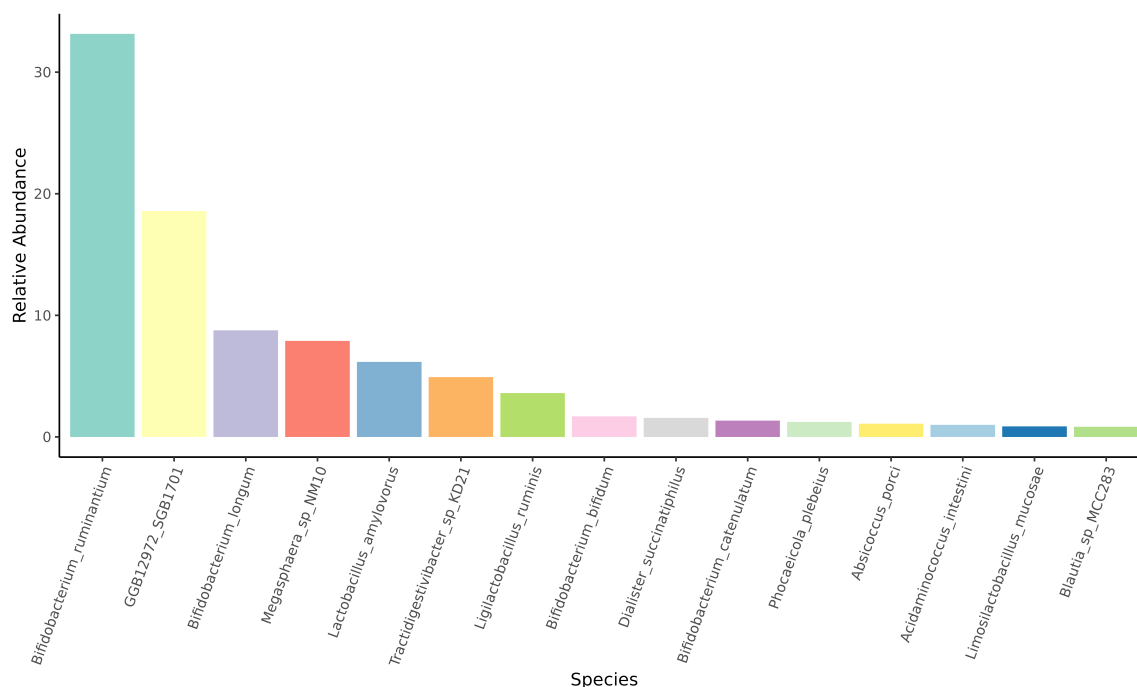
| Microbial Species Richness                                                                            | Description                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Score= 108</p>   | <p>Richness is the total number of microbial species identified within your sample. The number of species found in your gut microbiome is 108 and it is within the average Indian population range.</p> |
| Evenness Index                                                                                        | Description                                                                                                                                                                                             |
|  <p>Score= 0.51</p> | <p>Evenness describes how uniformly all the species are present in your gut microbiome. The value of species evenness in your sample is 0.51 and it is within the average Indian population range.</p>  |
| Microbiome Diversity Index                                                                            | Description                                                                                                                                                                                             |
|  <p>Score= 2.4</p> | <p>The microbiome diversity index is calculated using both richness and evenness. The value of microbiome index of your sample is 2.40 and it is within the average Indian population range.</p>        |



## Distribution of Species:

The top 15 species in your gut microbiome based on the relative abundance.\*

\*When a bacterial species has numbers in its name, it signifies a recent discovery that has not been formally named yet.



## Probiotic Species:

Probiotic species are microorganisms present in your gut microbiome that contribute in maintaining a good health supporting digestion, modulating the immune system, protecting the intestinal barrier and reducing inflammation. They also make the gut a hospitable place for the survival of different types of microorganisms. Some of these organisms are also known producers of Short Chain Fatty Acids (SCFAs). SCFAs are fatty acids that are produced by the gut microbiota during the fermentation of dietary fibers. SCFAs improve gut barrier function, possess anti-inflammatory and immunomodulatory properties, which help promote overall gut health and lower the risk of gastrointestinal diseases. (Xiong et al., "Health benefits and side effects of short-chain fatty acids", *Foods*, 2022, Vol. 11, Issue 18, pp. 2863)

This section presents the probiotic species detected in your gut microbiome and SCFA produced by them.

| Species                             | Presence | Short Chain Fatty Acid/s Produced | Description                                                                                                              |
|-------------------------------------|----------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <i>Alistipes putredinis</i>         | +        | Acetate, Propionate, Butyrate     | It is a bacteria from the phylum Bacteroidetes. It has been shown to have anti-inflammatory properties.                  |
| <i>Anaerostipes hadrus</i>          | -        | Not available                     | Not available                                                                                                            |
| <i>Bifidobacterium adolescentis</i> | -        | Acetate, Propionate, Lactate      | It is shown to have anti-inflammatory and anti-oxidant properties. It is also used as a probiotic to restore gut health. |

|                                        |   |                             |                                                                                                                                                                                           |
|----------------------------------------|---|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Bifidobacterium bifidum</i>         | + | Acetate, Lactate            | It is a bacteria known for their ability to ferment carbohydrates, help with digestion, nutrient absorption, and fighting off "bad" organisms that can cause disease.                     |
| <i>Bifidobacterium breve</i>           | + | Acetate, Lactate            | It is shown to have antimicrobial activity against human pathogens and immuno-stimulating properties.                                                                                     |
| <i>Bifidobacterium catenulatum</i>     | + | Acetate                     | It is shown to inhibit harmful pathogens, contribute to the maintenance of a balanced microbiome and support overall gut health and immune function.                                      |
| <i>Bifidobacterium longum</i>          | + | Acetate, Lactate            | It is shown to have anti-inflammatory properties, plays role in immunoregulation, and decreasing the infection risk. It is also used as a commercial probiotic to restore gut health.     |
| <i>Blautia wexlerae</i>                | + | Not available               | It is shown to alter energy metabolism of the host and anti-inflammatory properties. It is also shown to change the composition of the gut microbiome.                                    |
| <i>Collinsella aerofaciens</i>         | - | Not available               | Not available                                                                                                                                                                             |
| <i>Coprococcus catus</i>               | + | Butyrate, Acetate           | It is a bacteria that plays an important role in fermenting dietary fibers and producing propionate, butyrate and supports gut health.                                                    |
| <i>Dorea longicatena</i>               | - | Not available               | Not available                                                                                                                                                                             |
| <i>Faecalibacterium prausnitzii</i>    | + | Butyrate, Acetate           | It is one of the most abundant bacterial species found in the gut. It is known for its anti-inflammatory properties and importance in gut health.                                         |
| <i>Fusicatenibacter saccharivorans</i> | + | Lactate, Acetate, Succinate | It is bacterial species shown to have the ability to reduce inflammation, improve colitis symptoms, and restore gut microbiota balance in patients with ulcerative colitis.               |
| <i>Lactobacillus ruminis</i>           | - | Not available               | It inhibits harmful pathogens, enhances carbohydrate utilization for energy, and helps maintain a balanced microbial community, thereby supporting digestion and overall immune function. |
| <i>Megasphaera elsdenii</i>            | - | Not available               | Not available                                                                                                                                                                             |

|                               |   |                               |                                                                                                                                                                                              |
|-------------------------------|---|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Prevotella copri</i>       | - | Butyrate                      | It produces short-chain fatty acids (SCFAs) that protect the mucosal barrier and reduce inflammation. It has been associated with reduction of visceral fat and improved glucose metabolism. |
| <i>Roseburia faecis</i>       | + | Butyrate                      | It is shown to modulate immune response, reduce inflammation, improve symptoms of irritable bowel syndrome (IBS), and enhance the overall composition of the gut microbiota.                 |
| <i>Roseburia hominis</i>      | - | Butyrate                      | It is a bacteria known to produce beneficial short-chain fatty acids (SCFAs) and it has been shown to reduce neuroinflammation.                                                              |
| <i>Roseburia intestinalis</i> | - | Butyrate                      | It is shown to prevent intestinal inflammation and maintain energy homeostasis. It is also considered as a potential probiotic for treating gut associated disorders.                        |
| <i>Ruminococcus bromii</i>    | + | Acetate, Propionate, Butyrate | It is bacteria which breaks down complex carbohydrates, producing short-chain fatty acids important for colon health. It also helps promote gut microbiome balance.                          |

Among the Top 16 probiotic species predominantly observed in the Indian population, we identified 11 species in your gut microbiome.\*

**\*Note:** The reference ranges for the interpretation of the results are based on the analysis of more than 2500 gut microbiome samples collected across India in Phenome India: CSIR Cohort Health Knowledgebase.



# PROBIOTIC FOODS

## FRESH FOOD, HEALTHY GUT

Probiotics are food or supplements that contain living microorganisms (typically bacteria or yeast) that, when ingested in effective doses, provide health benefits to the host.

Trust  
YOUR  
Gut



**YOGURT** Probiotics in yogurt help maintain or restore the natural balance of gut microbiota, especially after disturbances caused by antibiotics or illness. Probiotics in yogurt can help prevent or treat diarrhea caused by infections or antibiotics.



**BUTTERMILK** is a traditional fermented dairy product, is characterized by a predominance of Firmicutes bacteria (96.2%). Within the Firmicutes phylum, the prominent genera include Lactobacillus, Streptococcus, and Enterococcus.

**APPLE-CIDER VINEGAR** may help prevent obesity and related metabolic issues like insulin resistance and fatty liver.



Consumption of **GREEN BANANAS** has been found to increase the presence of beneficial gut bacteria species that support essential biological processes.



**KIMCHI** is a traditional Korean fermented vegetable dish that is rich in probiotics, specifically lactic acid bacteria (LAB) such as Lactobacillus, Leuconostoc, and Pediococcus. These bacteria can help maintain a balanced gut microbiota by suppressing the growth of harmful bacteria. Regular consumption of kimchi can help maintain a diverse and healthy gut microbiota, which is essential for overall health



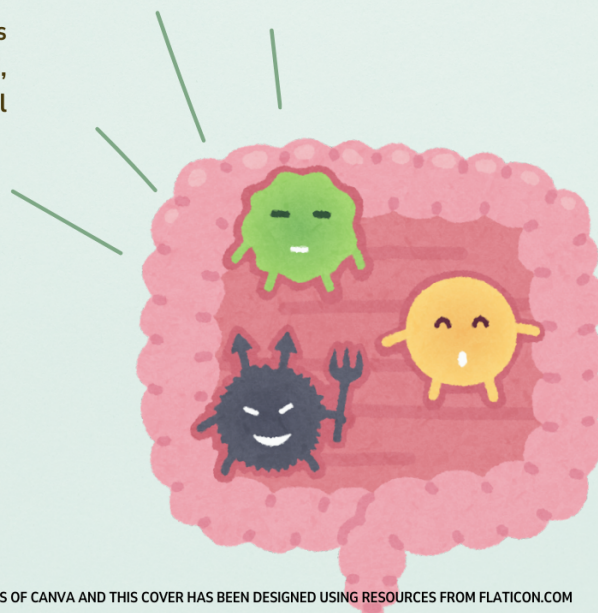
Consuming 85% **DARK CHOCOLATE** daily promotes the growth of Bifidobacteria and Lactobacilli, enhancing the diversity of gut microbial communities.



**OLIVE** and olive oil are often highlighted for their health benefits, including their potential impact on gut health and the microbiota. The connection between olives and gut health can be attributed to their high content of polyphenols, dietary fiber, and healthy fats, which can positively influence the gut microbiome.



**Almond:** Consuming almond and almond skins can influence the activities of intestinal bacteria, potentially acting as prebiotics.







Pickled vegetables can positively impact gut health due to their probiotic content. Pickled vegetables from pickled vegetables can introduce beneficial bacteria into the gut, promoting a diverse microbiota.



Consuming GARLIC can promote the growth of Lactobacillus acidophilus bacteria, potentially enhancing the probiotic benefits of products containing it. This could lead to improved gut health and strengthened immune function. Contains 'Inulin' and is used in cooking for its flavour and health benefits.



HERBAL TEA polyphenols from herbal tea are taken up by the gut microbiota and are recognized for their expanding role in contemporary nutrition.

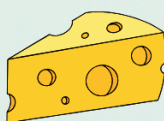
ONION decreases pH, boost butyrate production, and change enzyme activities in the gut microbiota. It is a common source of 'Inulin', found in various culinary dishes.



BERRIES have been shown to influence the balance of gut bacteria by decreasing the ratio of Firmicutes to Bacteroidetes (F/B) and increasing the presence of beneficial bacteria that promote health.



FISH-derived omega-3 fatty acids (n-3 PUFAs), along with probiotics or prebiotics, show promise as therapies for inflammatory bowel disease (IBD) and retinal diseases.



Consuming CHEESE as a probiotic can positively influence gut health and microbiota. Cheese contain beneficial live microorganisms, such as Lactobacillus and Bifidobacterium species, which can survive the gastrointestinal tract's harsh conditions.



GINGER shows tendency toward an increased Firmicutes-to-Bacteroidetes ratio, Proteobacteria and anti-inflammatory Faecalibacterium were found



MICROALGAE enhance the production of short-chain fatty acids (SCFAs), particularly acetate, butyrate, and propionate, during fermentation by colonic microbiota.

KOMBUCHA offers nutrition, boosts metabolism, and supports gut health by regulating the microbiome. This beverage serves as a dietary probiotic supplement rich in bioactive compounds and antioxidants, with measurable benefits for intestinal flora.



# Enhancing Gut Health

1. Regular exercise in daily routine can increase beneficial microbial species, enhance microflora diversity, and promote the growth of friendly bacteria. Even with a high-fat diet, exercise can potentially improve health outcomes by modifying and protecting the gut microbiota and help maintain intestinal health and integrity. It also offers protective benefits by lowering the risk of colon cancer, diverticulosis, and inflammatory bowel disease.

## Exercises for Healthy Digestive System:

**a. Walking:** Walking a few blocks daily can reduce bloating and gas, and promote regular bowel movements. Faster walking speeds and slower declines in walking speed increase gut microbiome diversity and beneficial bacteria like *Paraprevotella*, *Fusicatenibacter*, and *Alistipes*, enhancing gut health through anti-inflammatory properties and short-chain fatty acid production



**b. Yoga:** Yoga has been shown to lower stress levels, which can have a positive effect on your digestive system. Practicing specific yoga poses such as downward dog, boat, child's pose, upward dog, and triangle can support your aim for better digestion. Yogic practices like Dhauti, Vasti improves the variety of microbiome, which may be positively affects both the gut and mind.

2. **ANTIBIOTICS** are mainly prescribed for bacterial infections. They work by targeting harmful bacteria in the gut, helping to eliminate pathogens responsible for illnesses like urinary tract infections, strep throat, and pneumonia.



It's important to use antibiotics carefully and with medical guidance to avoid disrupting the natural balance of gut bacteria, which can lead to side effects like diarrhea, yeast infections, or antibiotic resistance. This approach helps minimize risks and ensures antibiotics are used effectively for their intended purpose.

3. **SMOKING** is associated with higher chances of developing gut disorders like inflammatory bowel disease (IBD), Crohn's disease, and ulcerative colitis. These conditions involve ongoing inflammation in the digestive tract, which can cause symptoms such as stomach pain, diarrhea, and rectal bleeding.



4. **DRINKING:** Regularly consuming alcohol over time is linked to higher chances of developing digestive issues like gastritis, pancreatitis, and alcoholic liver disease. These conditions can harm the digestive system and reduce its effectiveness.



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